Roberta Rocca

Research Scientist @ Google, London
+45 93 92 03 66 | rbrrcc@gmail.com | https://rbroc.github.io | rbroc

Work Experience

Research Scientist 05/2025 – Present
Google, Paradigms of Intelligence Team *London, United Kingdom*
Assistant Professor 12/2022 – 04/2025
Department of Culture, Cognition, and Computation, Aarhus University *Aarhus, Denmark*
Postdoctoral Researcher 03/2022 – 12/2022
Department of Culture, Cognition, and Computation, Aarhus University *Aarhus, Denmark*
Predictive Analytics Data Fellow 06/2021 – 08/2021
Centre for Humanitarian Data, United Nations *The Hague, The Netherlands (Remote)*
Postdoctoral Researcher 11/2019 – 03/2022
Psychoinformatics Lab, University of Texas at Austin *Austin, TX*

Education

PhD in Cognitive Science 09/2016 – 10/2019
Department of Linguistics, Cognitive Science, and Semiotics, Aarhus University *Aarhus, Denmark*
MSc in Cognitive Science 09/2014 – 07/2016
Centre for Mind/Brain Sciences, University of Trento *Trento, Italy*
BA in Philosophy 10/2010 – 03/2014
School of Humanities, University of Palermo *Palermo, Italy*

Additional Research Experience

Visiting Researcher 06/2024 – 07/2024
MilaNLP group, Department of Computing Sciences, Bocconi University *Milan, Italy*
Visiting Researcher 08/2018 – 03/2019
Department of Applied Mathematics and Computer Science, Technical University of Denmark *Copenhagen, Denmark*
Visiting Researcher 06/2019 – 07/2019
Department of Psychology, University of East Anglia *Norwich, United Kingdom*
Visiting Researcher 01/2018 and 10/2018
Institute of Cognitive Sciences and Technologies, National Research Council *Rome, Italy*
Research Assistant 03/2016 – 07/2016
Department of Psychology, University College London *London, United Kingdom*
Research Assistant 08/2015 – 03/2016
Department of Language and Cognition, Max Planck Institute for Psycholinguistics *Nijmegen, The Netherlands*

Teaching and Supervision

Data Science, Prediction, and Forecasting | *MSc Cognitive Science, Aarhus University* Spring 2024
Natural Language Processing | *MSc Cognitive Science, Aarhus University* Fall 2023
Applied Cognitive Science | *BSc Cognitive Science, Aarhus University* Spring 2023, Spring 2024
Cognitive Neuroscience [Teaching Assistant] | *BSc Cognitive Science, Aarhus University* Spring 2018
Cognitive Science | *MA Cognitive Semiotics, Aarhus University* Fall 2017
BSc (4x) and **MSc (3x) thesis supervision** | *Cognitive Science, Aarhus University* Fall 2023, Spring 2024
PhD co-supervision: Line Elgaard Kruse Danielsen | *Aarhus University* Fall 2021 – Spring 2024

Skills

Expertise: LLMs; Machine Learning; Cognition; Data Science; Statistics; Experimental Methods
Programming: Python, R, SQL, Matlab, bash, PERL, Git/GitHub, CI/CD, Docker
Main Libraries: Torch, TensorFlow/Keras, Transformers, Scikit-Learn, NumPy, SciPy, StatsModels, Pandas
Languages: English (fluent), Italian (native), Danish (fluent), French (intermediate)
Open Source: Co-developer of Turftopic, pliers, Neuroscout; contributor to MMTEB

Publications & Technical Reports (Selected)

Full list available on Google Scholar. Hyperlinks are included for each paper (hover over the title).

[forthcoming] **Rocca, R.**, Boukortt, S., Keeling, G., Street, W. (2026), Beyond Sally-Anne: Evaluating Theory of Mind in LLMs using Epistemic Schelling Points.

[under review] Kim, J., Street, W., **Rocca, R.**, Korngiebel, D. M., Waytz, A., Keeling, G., Evans, J. A., (2026), Inducing language models to assert their own consciousness restores human beliefs and values.

[under review] Meulemans, A., Wołczyk, M., Weis, A. M., Nasser, R., **Rocca, R.**, Kobayashi, S., Lajoie, G., Steger, A., Richards, B., Hutter, M., Manyika, J., Saurous, R. A., Sacramento, J. (2026), A game theory for foundation models shows new paths to rational cooperation through similarity inference.

[under review] **Rocca, R.**, Street, W., Keeling, G., Evans, J. A. (2026), Psychological coupling: the necessary science of human-AI interaction.

Rocca, R. & Tylén, K. (2026), Diversity and Interaction Structure Shape Performance and Search Dynamics in Joint Cognitive Search: An Agent-Based Simulation, *Proceedings of the Annual Meeting of the Cognitive Science Society (CogSci 2026)*

Enevoldsen, K., Chung, I., Kerboua, I., Kardós, M., . . . , **Rocca, R.**, . . . , & Muennighoff, N. (2025), MMTEB: Massive Multilingual Text Embedding Benchmark, *Proceedings of the International Conference on Learning Representations (ICLR 2025).*

Kardós, M., Enevoldsen, K., Kostkan, J., Kristensen-McLachlan, R. D., **Rocca, R.** (2025), Turftopic: Topic modelling with contextual representations from sentence transformers, *Journal of Open Source Software*

Kardós, M., Kostkan, J., Vermillet, A.Q., Nielbo, K., Enevoldsen, K., **Rocca, R.** (2025), S^3 – Semantic Signal Separation, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)*

Luo, X., Rechardt, A., Sun, . . . , **Rocca, R.**, . . . , & Love, B. C. (2025), Large language models surpass human experts in predicting neuroscience results, *Nature Human Behaviour*

Rocca, R., Pistilli, G., Maheshwari, K., Fusaroli, R. (2025), Introducing ELLIPS: an ethics-centered approach to research on LLM-based inference of psychiatric conditions, *Proceedings of the 2024 AAAI/ACM Conference on AI, Ethics, and Society (AIES 2024)*

Rocca, R. (2024), Interpreting R^2: a Narrative Guide for the Perplexed, *Towards Data Science*

Rocca, R., Lawall, K., Tsakiris, M., Cram, L. (2024), Communicating Europe: a computational analysis of the evolution of the European Commission's communication on Twitter. *Journal of Computational Social Science*, 1-52.

Coventry, K.R., Gudde, H. B., Diessel, H., Collier, J., . . . , **Rocca, R.**, . . . & Incel, O.D., (2023), Spatial communication systems across languages reflect universal action constraints. *Nature Human Behaviour*

Fusaroli, R., Weed, E., **Rocca, R.**, Fein, D., Naigles, L. (2023), Caregiver linguistic alignment to autistic and typically developing children: A natural language processing approach illuminates the interactive components of language development, *Cognition*

Hansen, L., **Rocca, R.**, Simonsen, A., Olsen, L., . . . & Fusaroli, R. (2023), Speech- and text-based classification of neuropsychiatric conditions in a multidiagnostic setting. *Nature Mental Health*

Rocca, R., Tamagnone, N., Fekih, S., Contla, X., Rekabsaz, N. (2023), Natural language processing for humanitarian action: challenges, opportunities, and the path towards a humanitarian NLP community, *Frontiers in Big Data*

de la Vega, A., **Rocca, R. (co-first)**, Blair, R., Mentch, J., Markiewicz, C., Ghosh, S., Poldrack, R., Yarkoni, T. (2022), Neuroscout: a unified platform for generalized and reproducible fMRI research, *eLife*

Rocca, R., Yarkoni, T. (2022), Language as a fingerprint: Self-supervised learning of user encodings using transformers, *Findings of the Association for Computational Linguistics: EMNLP 2022*

Rocca, R. (2021), Complex Systems Modeling for Humanitarian Action: Methods and Opportunities, *UN Centre for Humanitarian Data*

Rocca, R., Yarkoni, T. (2021). Putting psychology to the test: rethinking model evaluation through benchmarking and prediction, *Advances in Methods and Practices in Psychological Science*

Rocca, R., Coventry, K. R., Tylén, K., Staib, M., Lund, T. E., Wallentin, M. (2020). Language beyond the language system: dorsal visuospatial pathways support processing of demonstratives and spatial language during naturalistic fast fMRI, *NeuroImage*

Rocca, R., Wallentin, M., Vesper, C., Tylén, K. (2019). This is for you: social modulations of proximal vs. distal space in collaborative interactions, *Scientific Reports*

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